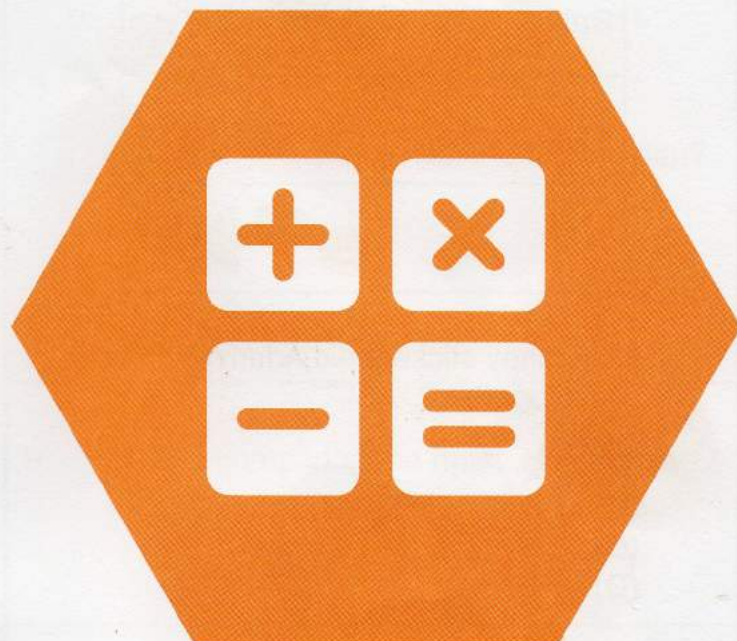




UNSW Global
THE UNIVERSITY OF NEW SOUTH WALES
SYDNEY • AUSTRALIA

PAPER B



2014 ICAS

International Competitions
and Assessments for Schools

MATHEMATICS

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DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

40 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.
Use a 2B or B pencil.
Do **NOT** use a pen.
Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.
Your score will be the number of correct answers.
Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).
Use the information provided to choose the **BEST** answer from the four possible options.
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a ruler and spare paper.
You are **NOT** allowed to use a calculator.

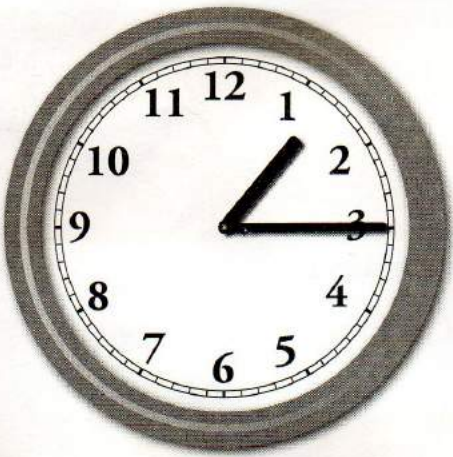
1. The table shows the stickers some students won this week.

Student name	Stickers
Adam	 
Rand	    
Sam	  
Tala	   

How many stickers did Adam and Rand win altogether?

- (A) 2
- (B) 3
- (C) 5
- (D) 7

2. What is the time on this clock?



- (A) 1:03
- (B) 1:15
- (C) 3:01
- (D) 3:07

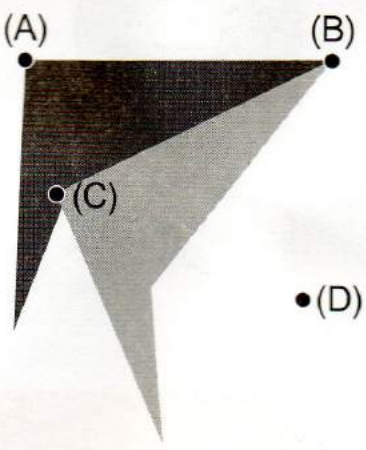
3. Mike had a bottle full of water. He poured all of the water into an empty glass. This filled the glass completely.

Which of these statements about the bottle and the glass MUST be true?

- (A) They have the same mass.
- (B) They have the same weight.
- (C) They have the same height.
- (D) They have the same capacity.

4. The picture shows two shapes.

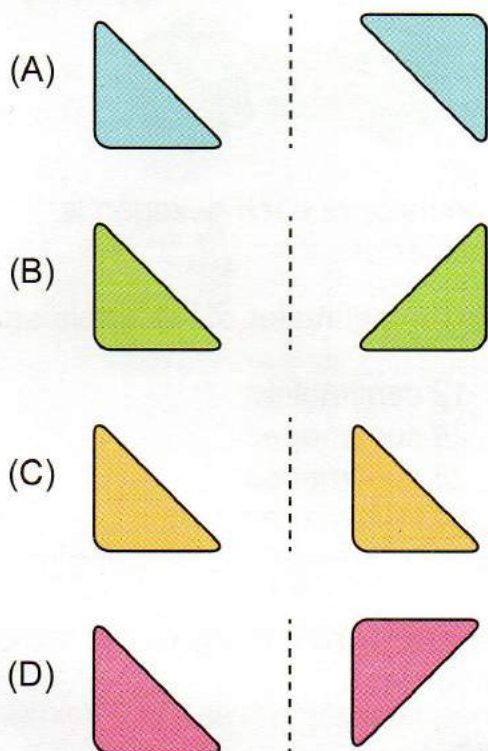
About which point can the darker shape be rotated to completely cover the lighter shape?



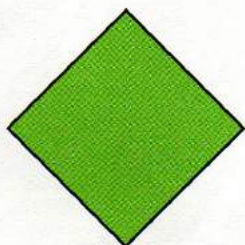
5. $8 + 11 - 7 =$?

- (A) 1
- (B) 4
- (C) 12
- (D) 26

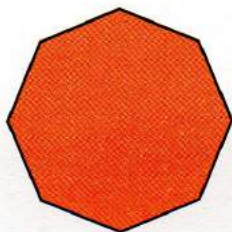
6. Which of these shows a shape reflected in the dotted line?



7. A square has two pairs of parallel sides.

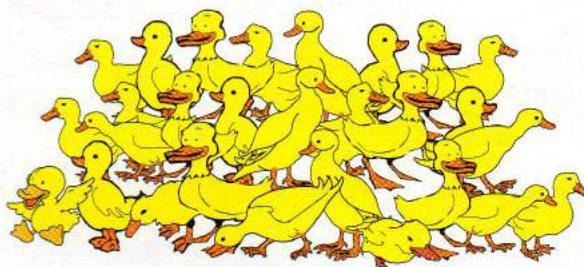


How many pairs of parallel sides does a regular octagon have?



- (A) 0
(B) 2
(C) 4
(D) 8

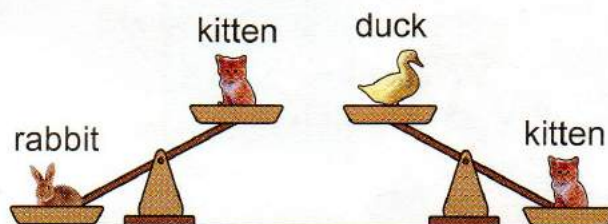
8. Tala had 28 ducklings in a pen. She took them out for a swim in groups of five until there were less than five left in the pen.



How many ducklings were left in the pen?

- (A) 3
(B) 4
(C) 5
(D) 23

9. Mike is comparing the mass of his three pets.



Which option lists Mike's pets in order from biggest mass to smallest mass?

- (A) duck, kitten, rabbit
(B) kitten, rabbit, duck
(C) rabbit, duck, kitten
(D) rabbit, kitten, duck

10. The table shows some information about the books some children read.

	Non-fiction	Fiction
Boys	20	16
Girls	5	15

How many children read fiction books?

- (A) 15
- (B) 16
- (C) 31
- (D) 56

11. A teacher dropped his keys in a measuring jug of water.



The level of the water rose from 4000 to 4100 on the scale on the jug.

What were the units of these measurements?

- (A) litres
- (B) millilitres
- (C) grams
- (D) kilograms

12. This shape has been made by joining two regular hexagons and one square.

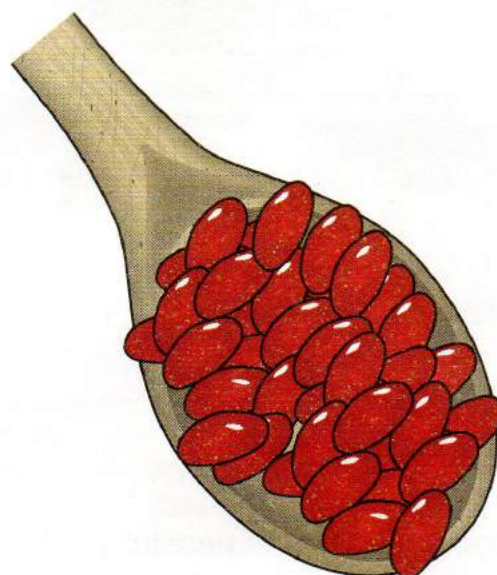


The perimeter of each hexagon is 12 centimetres.

What is the perimeter of the whole shape?

- (A) 12 centimetres
- (B) 24 centimetres
- (C) 28 centimetres
- (D) 32 centimetres

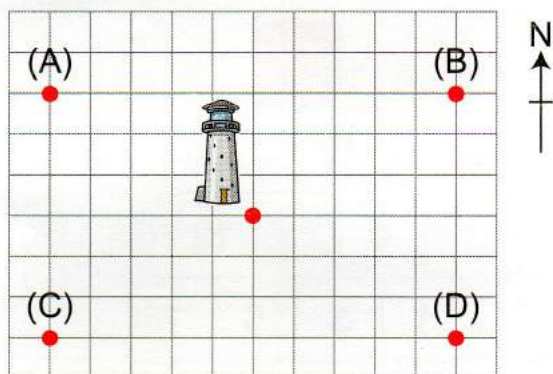
13. Approximately how many beans are on this spoon?



- (A) 4000
- (B) 100
- (C) 40
- (D) 10

14. Treasure is hidden at a point that is 3 units to the south and 5 units to the east of the lighthouse.

At which point is the treasure hidden?



15. Patrick found four different leaves.

Which leaf has the greatest perimeter?



(A)



(B)



(C)



(D)

16. Leila is helping her mum to bake cakes.

Each cake needs two cups of flour. For each cup of flour they need three eggs.

How many eggs does Leila need to make two cakes?

- (A) 3
- (B) 5
- (C) 6
- (D) 12

17. Natalia drew four shapes.



Shape 1



Shape 2



Shape 3

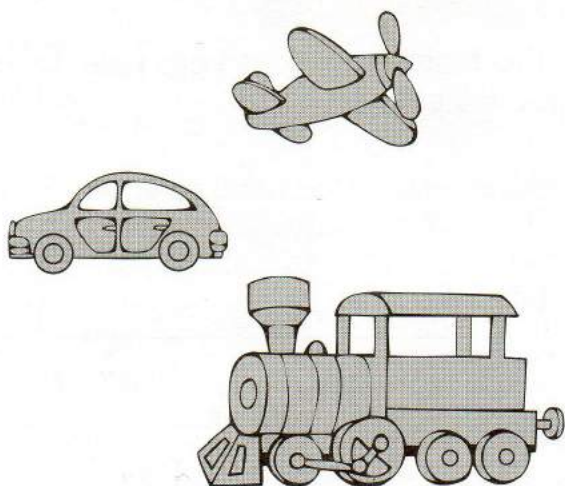


Shape 4

Which shapes have only one line of symmetry?

- (A) Shape 3 and Shape 4
- (B) Shape 2 and Shape 3
- (C) Shape 1 and Shape 3
- (D) Shape 1 and Shape 2

18. Max is painting some wooden toys. He can choose red, yellow or green. Each toy must be painted with a different paint.



How many different ways can Max paint his toys?

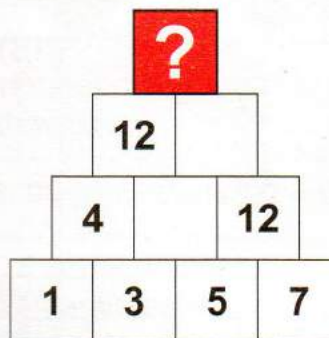
- (A) 12
- (B) 9
- (C) 6
- (D) 3

19. A school has 150 students. Two-thirds of the students are boys.

How many students are girls?

- (A) 50
- (B) 75
- (C) 100
- (D) 125

20. Sandy made a number pattern. She started on the bottom row. She added two numbers that were next to each other and wrote the answer in the box above. Some of the numbers are missing.



What must the value of ? be?

- (A) 22
- (B) 24
- (C) 32
- (D) 44

21. This fabric is being measured by a ruler that is marked in millimetres.



Which of these is closest to the length of the fabric?

- (A) 70 millimetres
- (B) 75 millimetres
- (C) 82 millimetres
- (D) 182 millimetres

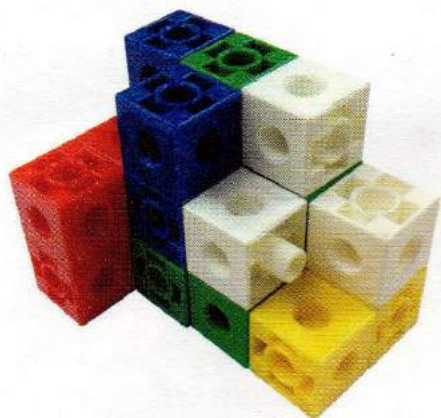
22. A local shop is offering one bouncy ball free with every five bouncy balls a customer pays for.

Jess pays the normal price for 30 bouncy balls.

How many bouncy balls will she receive in total?

- (A) 30
(B) 31
(C) 35
(D) 36

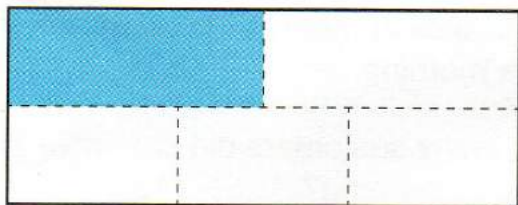
23. Sarah is making a solid cube using coloured blocks of the same size. She has completed one corner.



How many blocks has Sarah used so far?

- (A) 15
(B) 19
(C) 21
(D) 23

24.



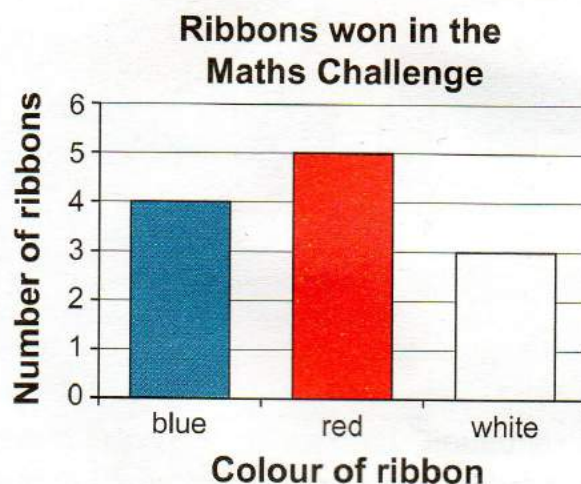
What fraction of this shape is shaded?

- (A) $\frac{1}{2}$ (B) $\frac{1}{4}$
(C) $\frac{1}{5}$ (D) $\frac{1}{3}$

25. At Sam's school, points and ribbons are awarded for the first three places in each round of a Maths Challenge. The table shows the point value for each colour ribbon.

Colour of ribbon	Number of points
blue	3
red	2
white	1

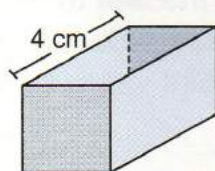
This graph shows the number of ribbons Year 4 won in the Maths Challenge.



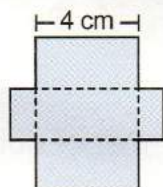
What was the total number of points won by Year 4?

- (A) 5
(B) 12
(C) 23
(D) 25

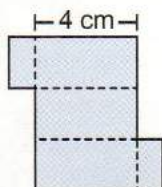
26. This picture shows a box without a lid.



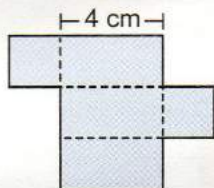
Which of these shows the net for this box?



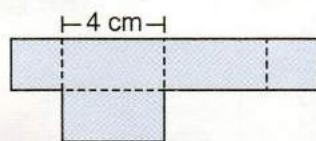
(A)



(B)

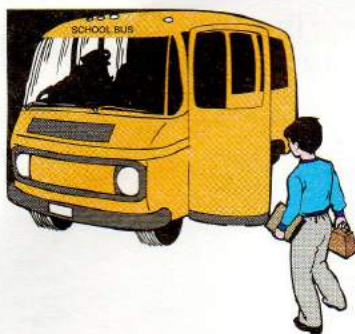


(C)



(D)

27. Billy usually catches the 8:25 am bus to school. The trip usually takes 20 minutes.



This morning the bus picked Billy up 5 minutes late. The trip took 7 minutes less than usual.

At what time did Billy get to school?

- (A) 8:57 am
- (B) 8:50 am
- (C) 8:33 am
- (D) 8:43 am

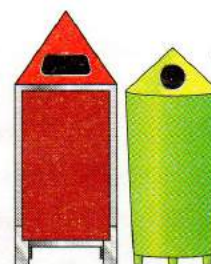
28. This picture shows two bins for waste when viewed from the south.



What do the bins look like viewed from the east?



(A)



(B)



(C)



(D)

29. Sonia has a shop.

On Tuesday she had 140 customers. This was double the number of customers that she had on Monday. On Monday she had 30 more customers in the afternoon than in the morning.

How many customers did she have on Monday morning?

- (A) 20
- (B) 40
- (C) 50
- (D) 70

30. Using the ruler in the picture, how much taller than Tala is Adam?



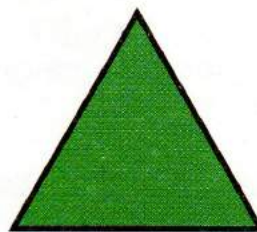
- (A) 2 centimetres
(B) 20 centimetres
(C) 2 metres
(D) 20 metres

31. George is saving money to buy a new skateboard. In the first month he saved \$12. Every month after the first he saved \$2 more than the month before.

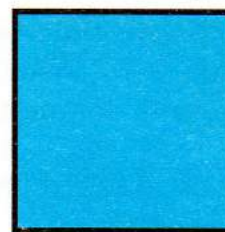
How much money did George save in total in 5 months?

- (A) \$20
(B) \$60
(C) \$70
(D) \$80

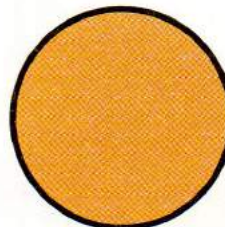
32. Which shape has NO lines of symmetry?



(A)



(B)

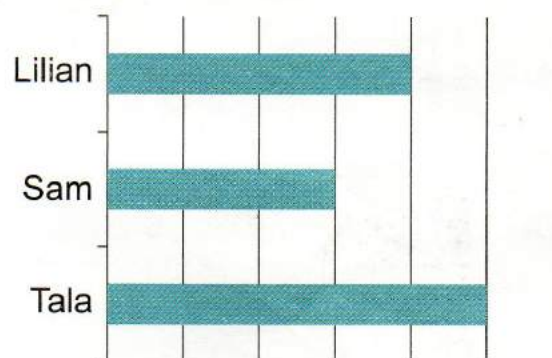


(C)



(D)

33. This graph shows the number of books read by three friends.



Tala read 10 books.

How many books did the three friends read altogether?

- (A) 24
(B) 17
(C) 14
(D) 12

34. Mike thinks of two numbers.

When he adds the two numbers, the result is 15. When he multiplies the two numbers, the result is 54.

Which of these is the smaller of the two numbers?

(A) 3
(B) 6
(C) 9
(D) 12

35. Ann needs exactly four-and-a-half cups of water to make soup.

She only has a glass jar that measures three-quarters of a cup.

How many times will she need to fill the glass jar to follow the recipe?

(A) $1\frac{1}{2}$ (B) 3
(C) $4\frac{1}{2}$ (D) 6

36. The two robots start on these squares.



They are programmed to follow these rules:

- $2R - 1$ means move two squares to the right and then one square down
- $3L + 1$ means move three squares to the left and then one square up

Which of these rules will result in the two robots moving to the same square?

		
(A)	$1R + 1$	$1L - 1$
(B)	$2R + 1$	$2L - 1$
(C)	$3R$	$3L$
(D)	$4R$	$2R - 1$

37. Mr Lee makes jewellery. He makes two different kinds of bracelets. The Supa bracelet has three star beads and four oval beads. The Glowy bracelet has five star beads and three oval beads.



Supa



Glowy

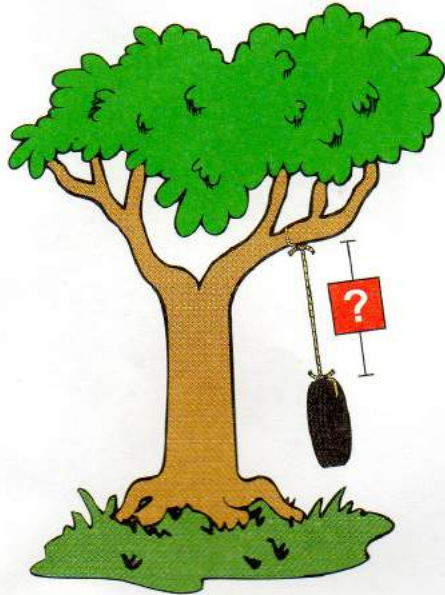
Mr Lee made 25 bracelets. Altogether he used 101 star beads.

How many Glowy bracelets did he make?

(A) 20
(B) 16
(C) 13
(D) 12

38. Nathan wanted to hang a tyre from a tree to make a swing.

He had 4 metres of rope. He used one-quarter of the rope to tie it to a strong branch of the tree. He used 75 centimetres to tie around the tyre.



What is the length of the rope between the tyre and the branch?

- (A) 225 centimetres
- (B) 300 centimetres
- (C) 325 centimetres
- (D) 500 centimetres

39. Melissa bought 8 pens and 5 notebooks. Each pen cost \$3.50.

Melissa spent \$2 less buying the pens than buying the notebooks.

What was the cost of each notebook?

- (A) \$1.50
- (B) \$5.20
- (C) \$5.50
- (D) \$6.00

40. Amber planned a party. Each child was asked to bring a cupcake, juice and a present to the party. They were to bring one of each type of item.



Each child brought at least one item.

- 16 children brought all three items.
- 5 children brought the cupcake and the present only.
- 1 child brought the cupcake only.
- 1 child brought the juice only.
- 3 children brought the cupcake and the juice only.

There were 25 cupcakes, 21 juices and 26 presents to share altogether.

How many children were at the party?

- (A) 26
- (B) 27
- (C) 31
- (D) 72

Acknowledgment

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The following year levels should sit THIS Paper:

Australia	Year 4
Brunei	Primary 4
Indonesia	Year 5
Malaysia	Standard 4
New Zealand	Year 5
Pacific Region	Year 5
Singapore	Primary 3
South Africa	Grade 4



PAPER
B

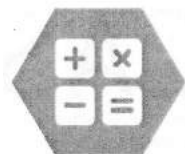


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M | 2014 Mathematics Answer Keys

ICAS

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Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Papers G & H	Papers I & J
1	B	D	B	D	A	A	B	A
2	B	B	C	A	A	A	C	B
3	C	D	B	A	D	C	B	C
4	B	B	D	C	B	D	A	B
5	C	C	C	A	C	D	C	C
6	A	B	A	C	D	B	C	A
7	C	C	D	A	A	C	A	A
8	B	A	C	B	D	A	D	B
9	C	D	A	D	A	B	C	C
10	D	C	D	B	D	B	A	B
11	B	B	C	D	D	A	B	C
12	A	B	B	A	B	B	D	D
13	C	C	D	D	D	B	A	D
14	C	D	D	C	B	C	D	A
15	B	C	B	C	B	C	B	B
16	A	D	D	D	C	B	A	B
17	D	A	A	B	B	C	D	C
18	D	C	B	A	A	B	B	D
19	B	A	D	A	B	C	B	D
20	B	C	A	D	C	D	A	D
21	C	B	C	B	B	C	A	B
22	B	D	D	A	D	A	C	A
23	C	C	C	C	C	A	C	C
24	A	B	A	C	B	B	B	A
25	D	D	B	B	C	D	C	B
26	C	C	B	D	D	C	D	A
27	B	D	C	C	C	B	B	D
28	B	A	A	C	C	A	D	A
29	A	A	D	D	A	D	B	D
30	D	B	C	B	A	C	B	B

(Please turn over)



M

2014 Mathematics Answer Keys

ICAS

International Competitions
and Assessments for Schools

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Papers G & H	Papers I & J
31	C	D	D	A	C	A	C	C
32	D	D	C	B	C	B	A	B
33	A	A	A	B	A	C	D	D
34	B	B	C	D	D	B	A	D
35	A	D	B	A	B	C	B	C
36	D	D	C	82 082	36 036	327	120	120
37	A	C	B	21 021	13 013	256	41 041	864
38	B	A	C	92 092	327	21 021	83 083	478
39	C	D	B	125	12 012	640	864	540
40	C	C	C	24 024	33 033	41 041	45 045	36 036

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